# Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

## PROFESSIONAL SUMMARY

Applied AI engineer who **built AgentProbe end-to-end** - the exact system Cohere's Agentic Workflows team builds: production LLM agents with multi-step reasoning (ReAct, Plan-and-Execute), RAG pipelines (ChromaDB), multi-provider routing (Anthropic, OpenAI, Groq, Ollama), and a 5-metric evaluation framework with CI/CD regression gates. Built from prototype to open-source - not just a demo. **LangChain + LangGraph** as the core orchestration stack; **8+ years production Python**; full-stack delivery across 120+ Agile sprints at Entrata (100K+ users, 99.5% uptime). AI on FHIR: HIPAA-compliant clinical NLP platform (BERT, spaCy, FHIR) - proof of building AI for domain-specific enterprise workflows. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).
*github.com/venkatalakshmiparimala/AgentProbe*

## CORE COMPETENCIES

Agentic Workflows (LangGraph, ReAct, Plan-Execute) | LLM Integration (Anthropic, OpenAI, Groq, Ollama) | RAG Pipelines & Vector Search (ChromaDB) | LLM Evaluation Framework (LLM-as-Judge) | Multi-Agent Orchestration | Prompt & Context Engineering | Python (8+ years production) | TypeScript / JavaScript / FastAPI | CI/CD Eval Gates (GitHub Actions) | AWS / Docker / Production at Scale | Clinical NLP (BERT, spaCy, FHIR) | Enterprise-Grade Reliability & Observability

## EXPERIENCE

### Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

**Software Engineer → Associate Software Engineer → Trainee**

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with careful output validation and quality monitoring.
- Built and delivered **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - startup pace with enterprise-grade uptime (99.5% over 6+ months).
- Led Docker-based containerization for 30%+ of microservices; improved production query performance by **30%**; built **5+ internal platform tools** adopted across squads, reducing resolution time by **40%**.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users.
- Drove secure coding standards and peer code review practices; mentored engineers across all major releases.

### George Mason University

Aug 2025 - May 2026

**Graduate Teaching Assistant - Python & Databases**

Fairfax, Virginia

- Mentored 30+ students; led technical discussions, designed exercises, and drove iterative quality improvement - the same communication and mentoring pattern Cohere's Applied AI role requires.

## PROJECTS

### AgentProbe: Production Agentic AI Platform - LLMs, RAG, Multi-Agent, Evaluation OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the same system Cohere's Applied AI team ships - production-grade LLM agents, from prototype to open-source. **Agentic workflows**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) using ReAct and Plan-and-Execute patterns - multi-step reasoning, tool calls, failure handling, and inter-agent state handoffs. **LLM integration**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type, latency, and capability. **RAG pipeline**: ChromaDB + embeddings + retrieval quality scoring across multi-step agent runs. **Evaluation framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with per-run regression tracking and CI/CD gates - agents must be reliable, observable, and auditable from day one. **Prompt & context engineering**: context window management, token budget optimization, short vs. long-term memory patterns. **Platform**: 15-endpoint FastAPI backend + React observability dashboard. *github.com/venkatalakshmiparimala/AgentProbe*

### AI on FHIR: Enterprise-Grade Clinical NLP - LLMs + BERT + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

AI system for clinical decision support in a regulated enterprise domain - same pattern Cohere applies to enterprise customer workflows. **Domain-specific agents**: NLP routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC**: enterprise-grade access controls built in from day one. **~60% retrieval time reduction.** Demonstrates building AI for domain-specific enterprise workflows with reliability and compliance requirements. *github.com/venkatalakshmiparimala/AI_on_FHIR*

### CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

## EDUCATION

**M.S. Data Analytics Engineering** - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

**B.Tech Computer Science & Engineering** - Sree Venkateshwara College of Engineering 2013-2017

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

**AI & Agents:** LangChain, LangGraph, Multi-Agent Orchestration, ReAct / Plan-and-Execute Patterns, LLM APIs (Anthropic, OpenAI, Groq, Ollama), Prompt Engineering, Context Engineering, Token Management, RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates

**NLP & ML:** BERT, spaCy, Hugging Face, PyTorch, scikit-learn, Embeddings, Feature Engineering, NumPy, Pandas, Statistical Modeling

**Cloud & DevOps:** AWS (S3, EC2, RDS, Glue), Azure, Docker, FastAPI, Flask, React, PostgreSQL, MySQL, RabbitMQ, CI/CD (GitHub Actions)

**Process:** TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum